



Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!) ○Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration Configure

Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` – installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements 1.

Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Student ID: 2024300753

Course/Year/Section: BSCS 3

Date Submitted: September 3, 2026

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS) Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu OS
Virtualization Software (VirtualBox/VMware)	Oracle Virtual Box
Ubuntu Server Version	Ubuntu-26.04-live-server-amd64
VM Name	TerranovaVMAkhylle
RAM Allocated	6 GB
Storage Allocated	100 GB
Network Mode	NAT
SSH Client Used	OpenBSD Secure Shell server

3 VM CONFIGURATION

Setting	Value
VM Name	TerranovaServerAkhylle
OS Type	Ubuntu (64-bit)
Memory	6000 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION Commands

Executed:

- `sudo systemctl status ssh`
- `sudo apt update`
- `sudo apt install openssh-server -y`
- `sudo systemctl enable ssh`
- `sudo systemctl start ssh`
- `ip a`

- ssh Terranova@192.168.137.113

Server IP Address: 192.168.137.113

Network Interface: Inter(R) Dual Band Wireless-AC 7625 #2

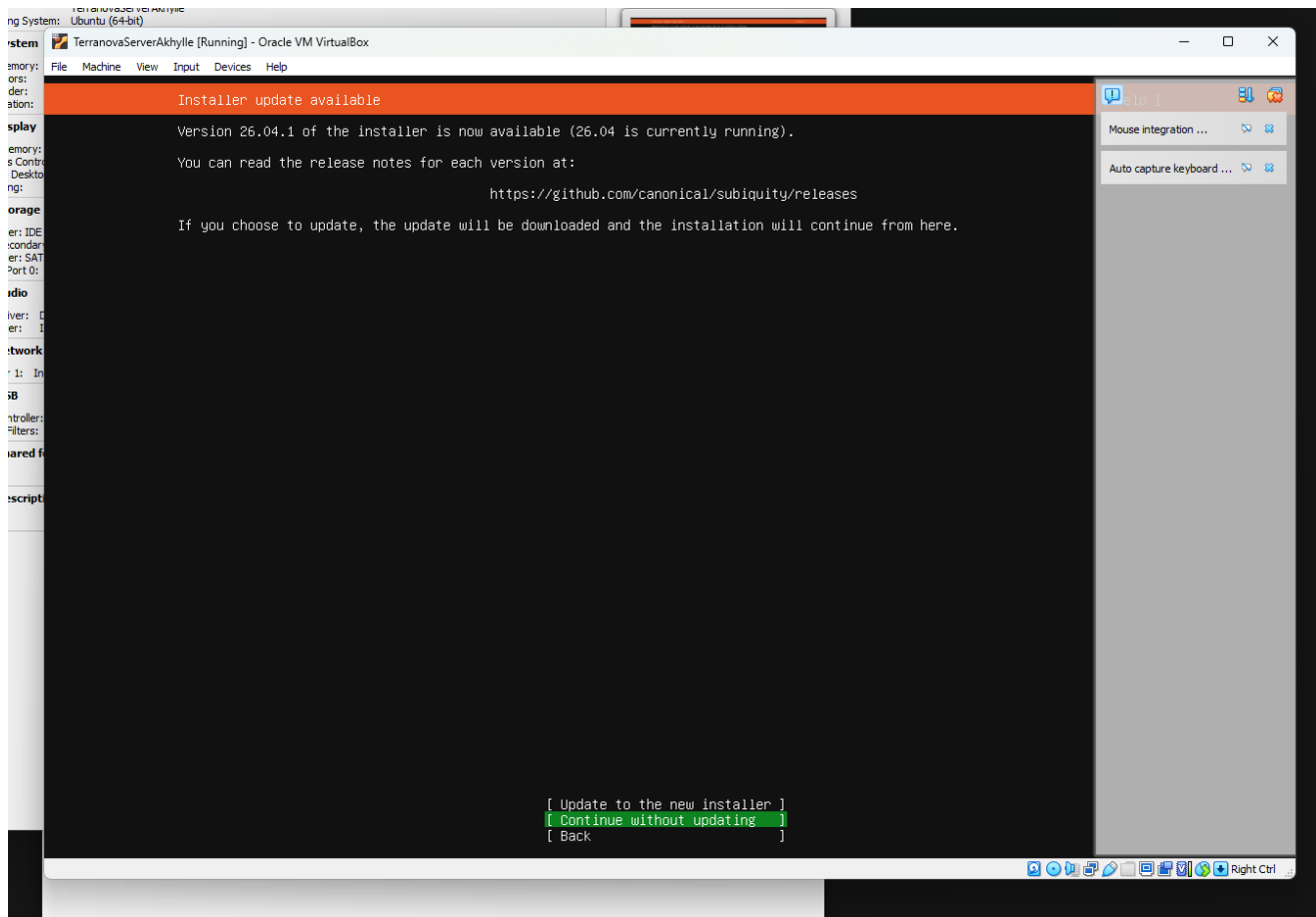
6. REMOTE CONNECTION Connection

Details:

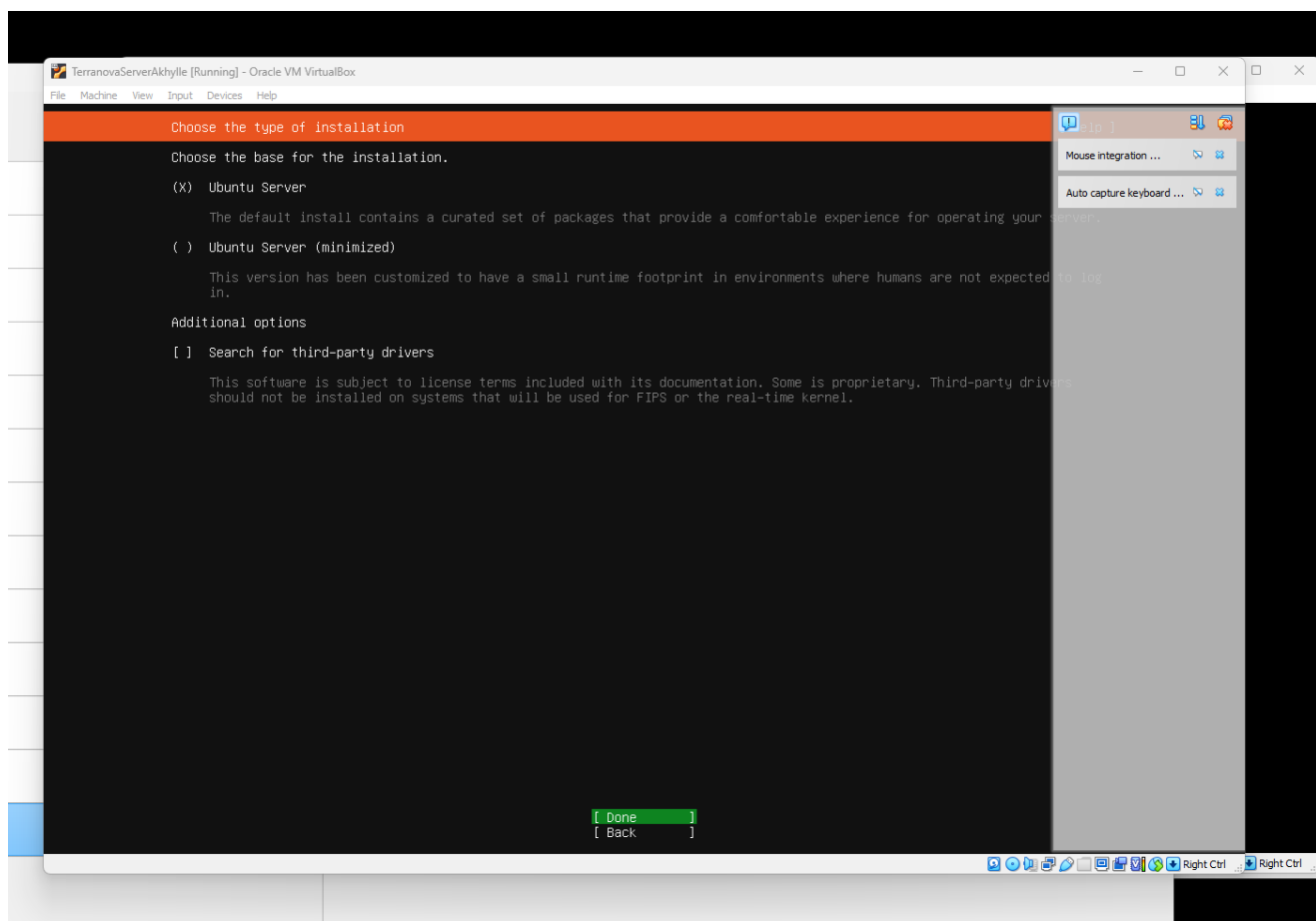
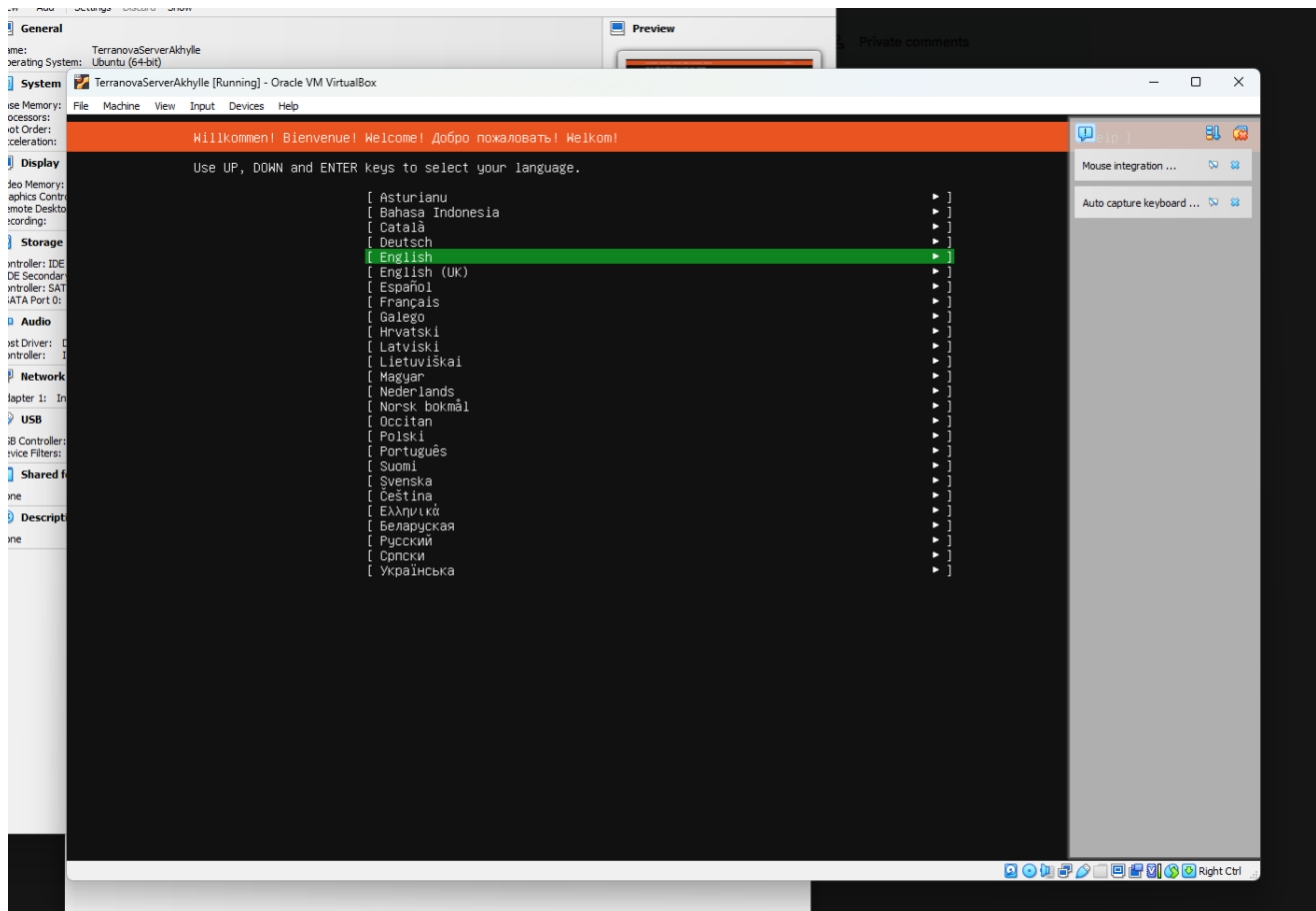
Parameter	Value
Server IP	192.168.137.113
Username	Terranova
SSH Port	24

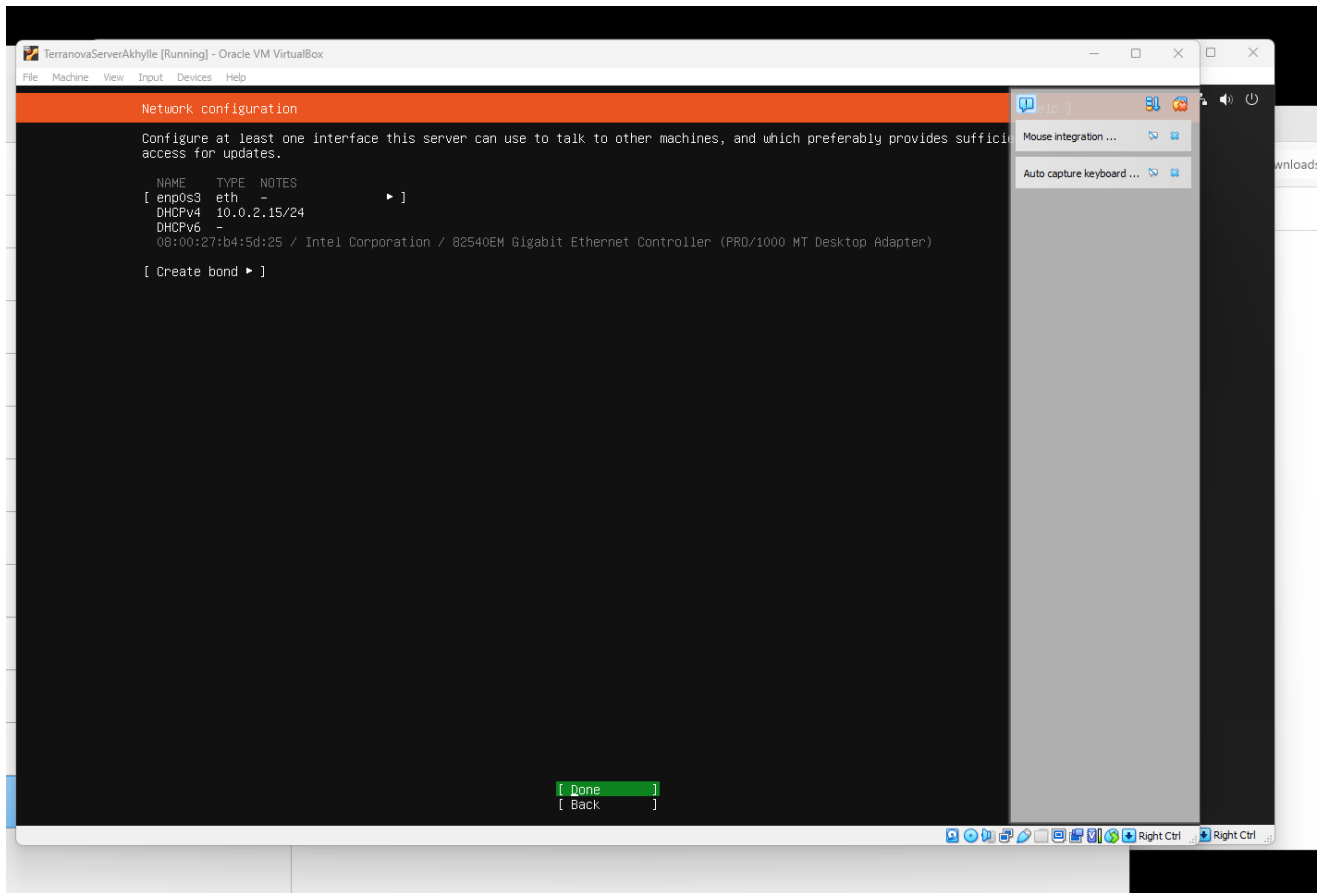
Screenshots

VM Server Installation

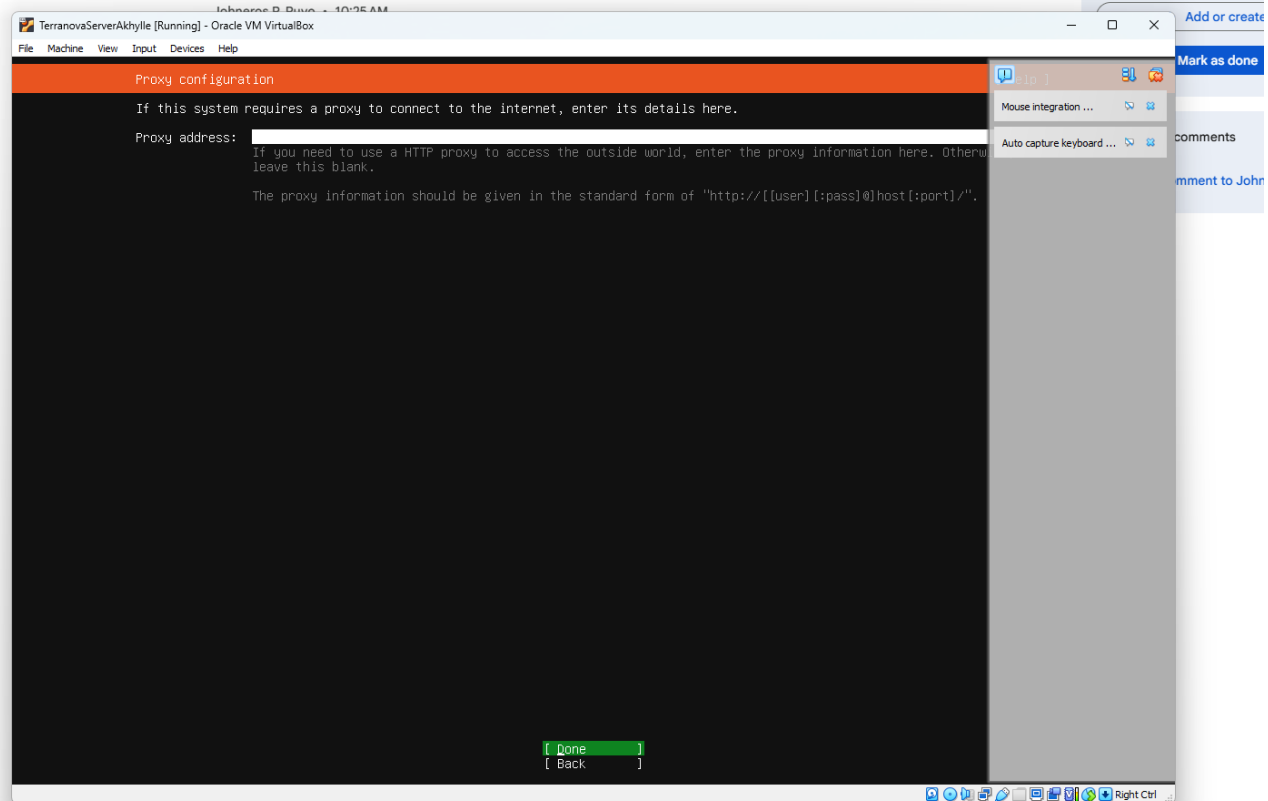


VM Server Prep





LAB 2 - VIRTUAL SERVER



LAB 2 - VIRTUAL SERVER

TerranovaServerAkhyle [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Ubuntu archive mirror configuration

If you use an alternative mirror for Ubuntu, enter its details here.

Mirror address:

You may provide an archive mirror to be used instead of the default.

This mirror location passed tests.

```
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Get:3 file:/cdrom/resolute Release.gpg [228 B]
Get:3 file:/cdrom/resolute Release.gpg [228 B]
Get:4 http://ph.archive.ubuntu.com/ubuntu/resolute InRelease [136 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu/resolute-updates InRelease [137 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu/resolute-backports InRelease [137 kB]
Fetched 409 kB in 2s (240 kB/s)
Reading package lists...
```

[Done]
[Back]

Mouse integration ...
Auto capture keyboard ...

Your work

Add or create

Mark as done

comments

comment to Johnner

LAB 2 - VIRTUAL SERVER

TerranovaServerAkhyle [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Guided storage configuration

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[VBOX_HARDDISK_VB0ae4bf14-55508b37 local disk 100.000G ▼]

(X) Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

[] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

() Custom storage layout

[Done]
[Back]

Mouse integration ...
Auto capture keyboard ...

Your work

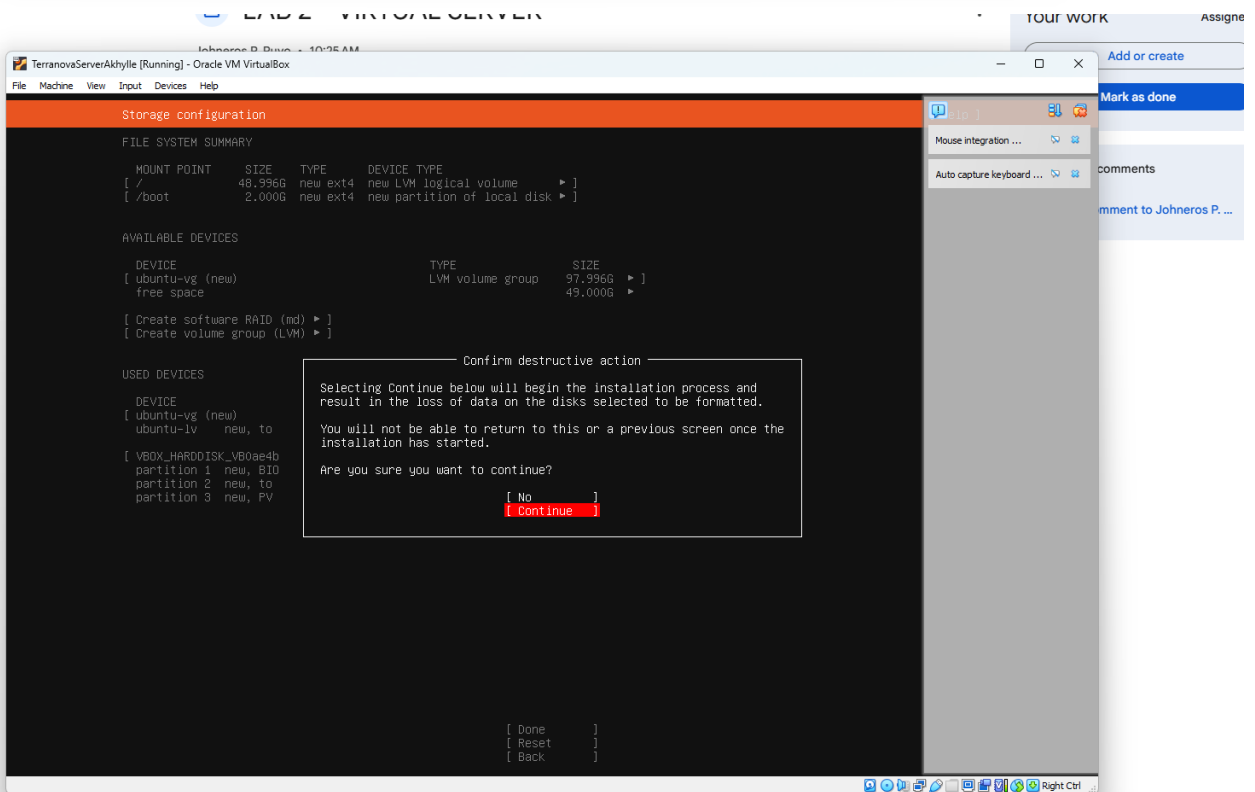
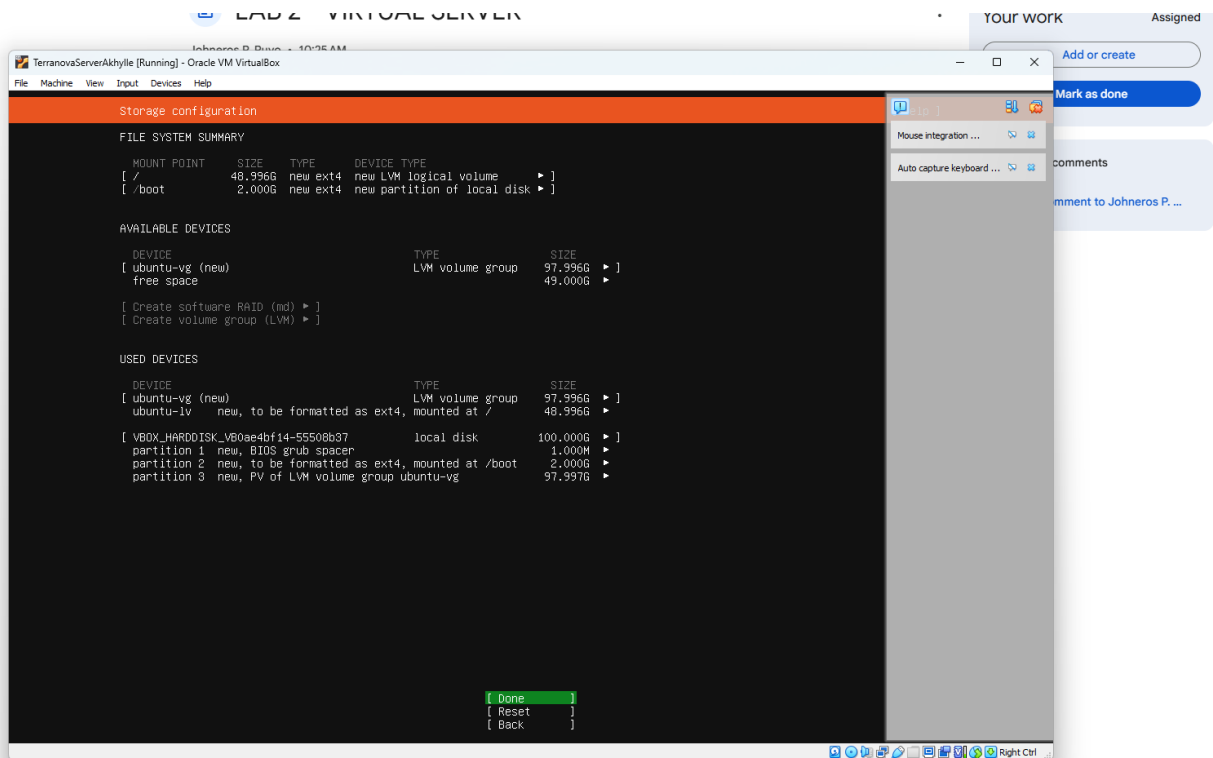
Assigned

Add or create

Mark as done

comments

comment to Johnner P. ...



Profile Configuration

TerranovaServerAkhille [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

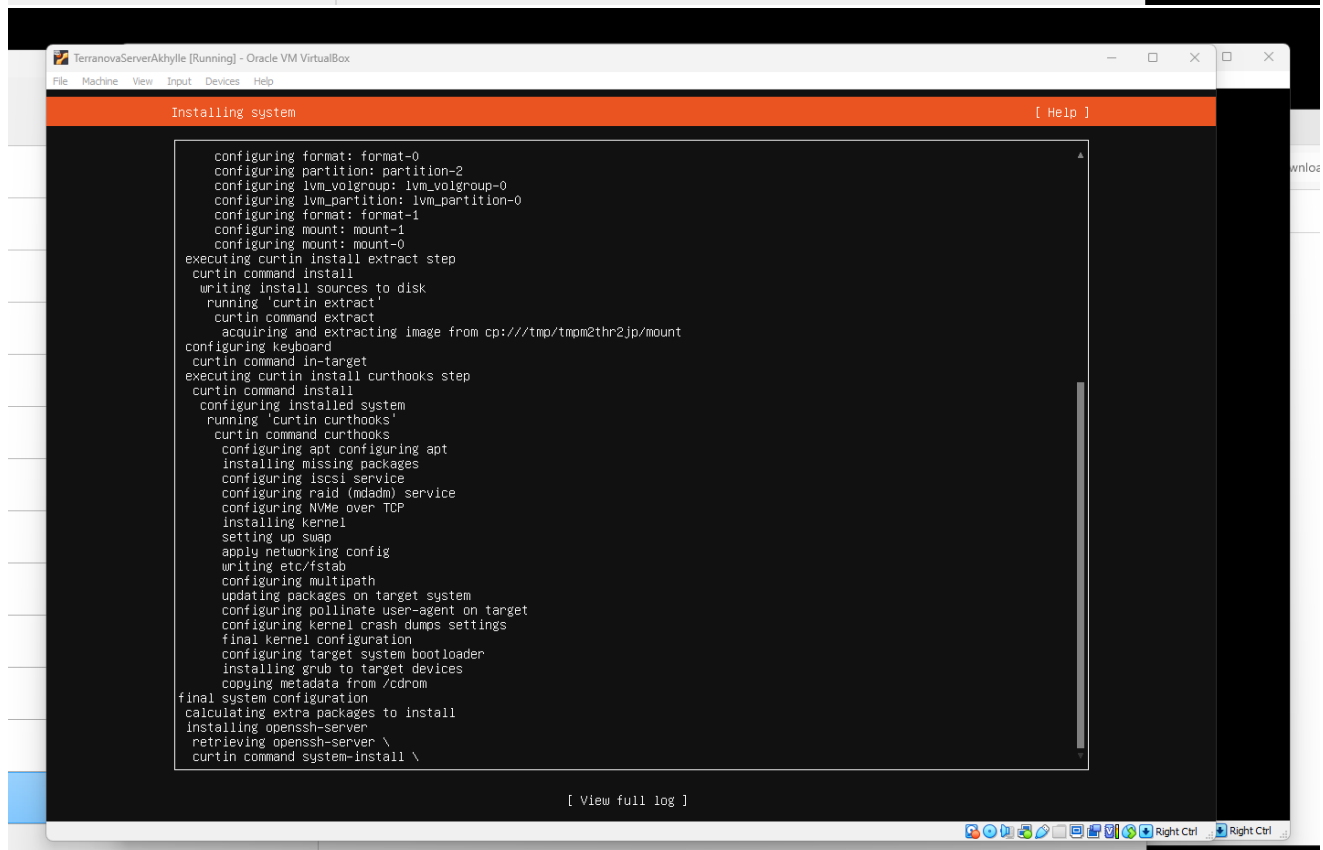
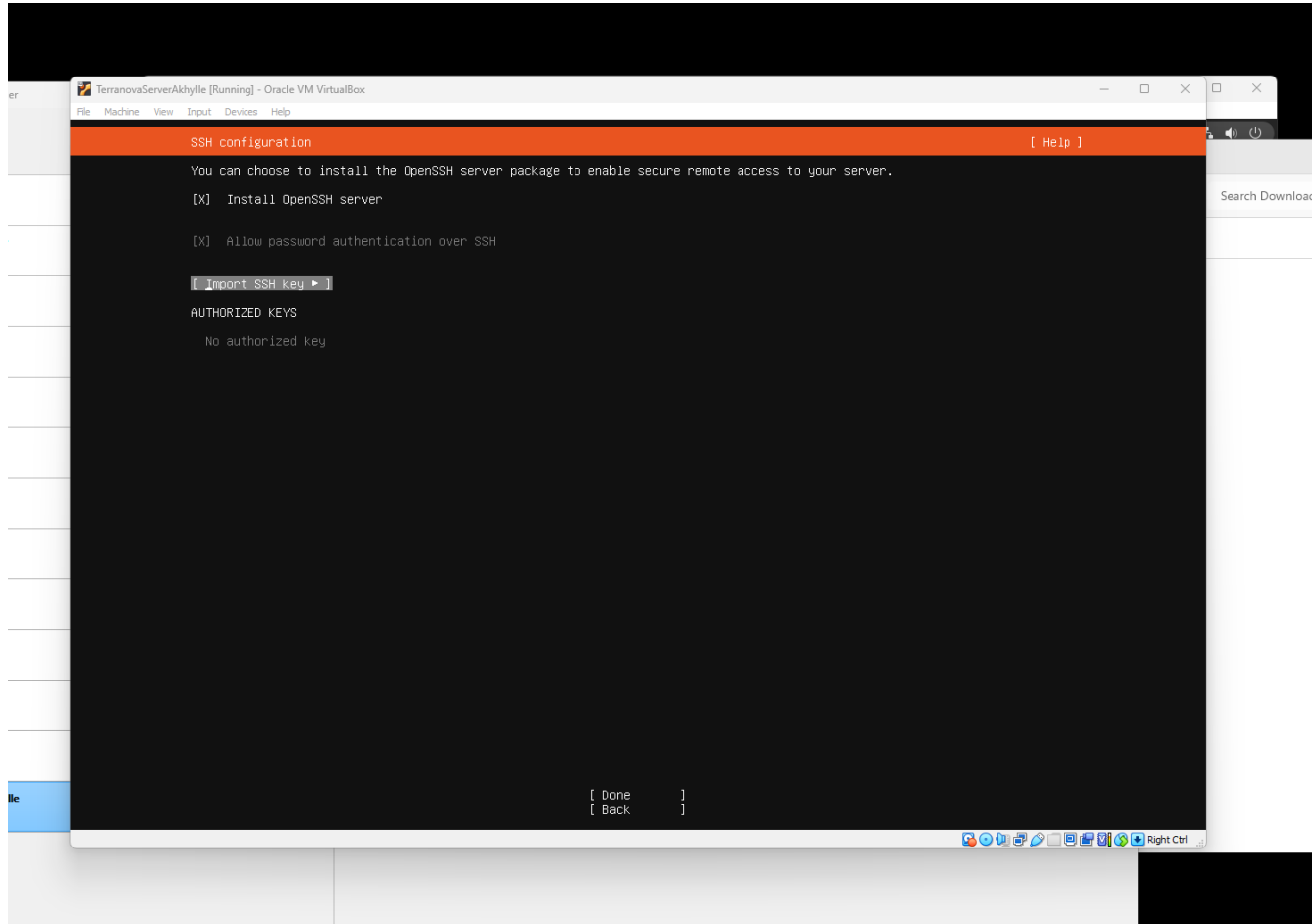
Choose a password:

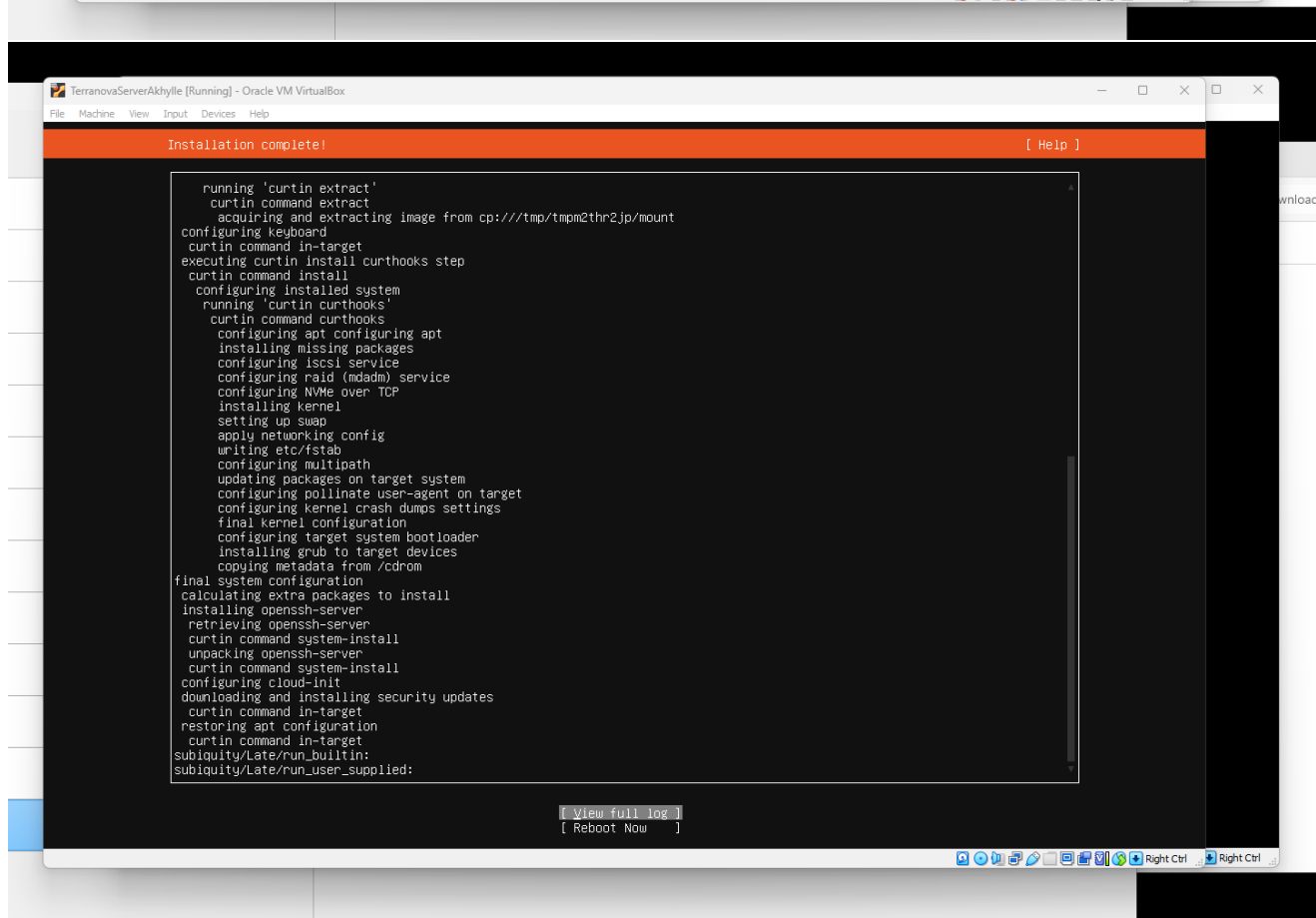
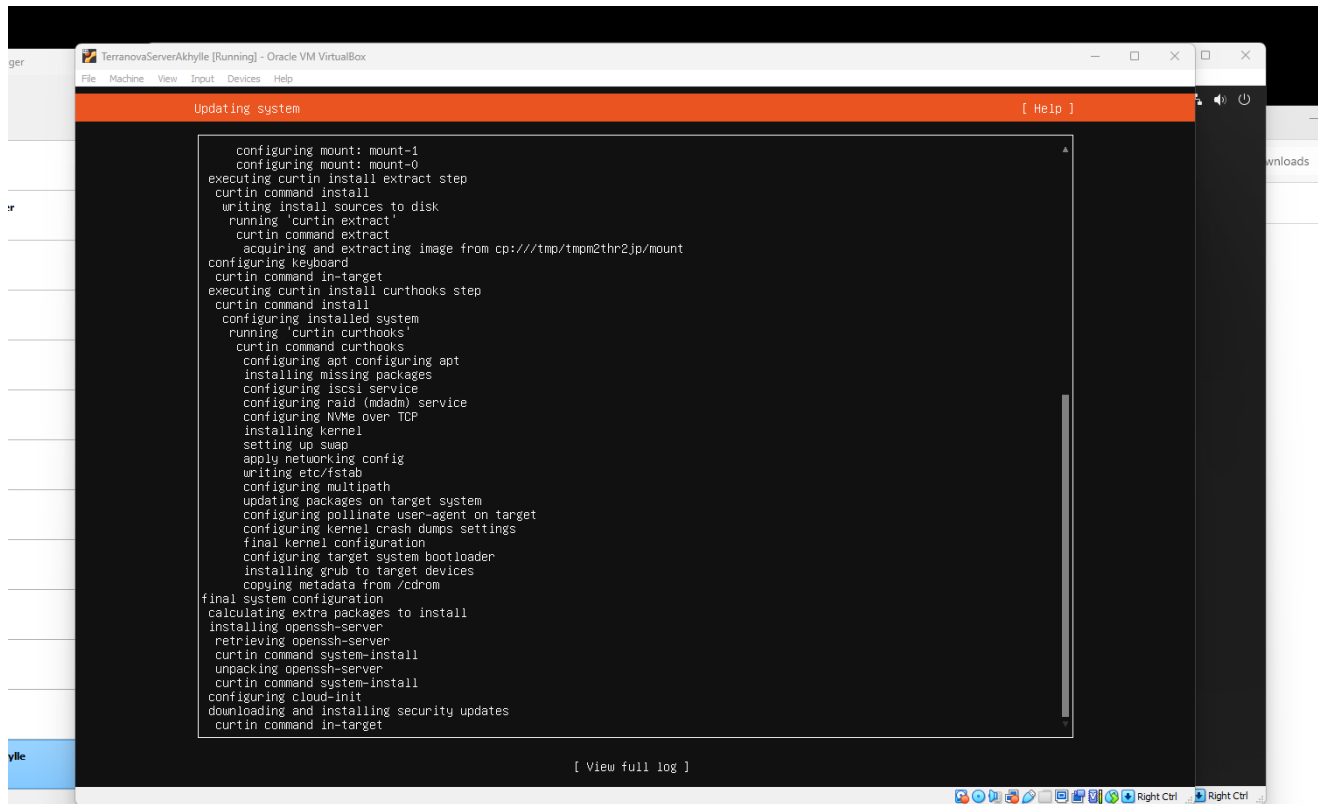
Confirm your password:

[Done]

Right Ctrl

Install SSH Server





Server Prep

```
terranova login: terranova
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 02:54:38 AM UTC 2026

System load:  0.12                Processes:    155
Usage of /:   14.9% of 47.9GB     Users logged in: 0
Memory usage: 4%                 IPv4 address for enp0s3: 10.0.2.15
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

terranova@terranova:~$
```

```
TerranovaServer@kyle (Running) - Oracle VM VirtualBox
File  Machine  View  Input  Devices  Help

Login incorrect

terranova login: terranova
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

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applicable law.

terranova@terranova:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:27:14:5d:25 brd ff:ff:ff:ff:ff:ff
    altname enp0s3v4-SGCS
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86107sec preferred_lft 86107sec
    inet6 fe80::a00:2714:5d25:64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
terranova@terranova:~$
```

```
terranova@terranova:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 02:53:14 UTC; 22min ago
   Invocation: c47838c2136a4655be68bc9c2bafacb5
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1369 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1372 (sshd)
     Tasks: 1 (limit: 3768)
    Memory: 1.6M (peak: 2.6M)
       CPU: 17ms
   CGroup: /system.slice/ssh.service
           └─1372 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 02:53:14 terranova systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 03 02:53:14 terranova sshd[1372]: Server listening on 0.0.0.0 port 22.
Sep 03 02:53:14 terranova sshd[1372]: Server listening on :: port 22.
Sep 03 02:53:14 terranova systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
terranova@terranova:~$
```

```

Sep 03 02:53:14 terranova systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
terranova@terranova:~$ sudo apt update
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Err:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
58 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Temporary failure resolving 'security.ubuntu.com'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
terranova@terranova:~$
```

```

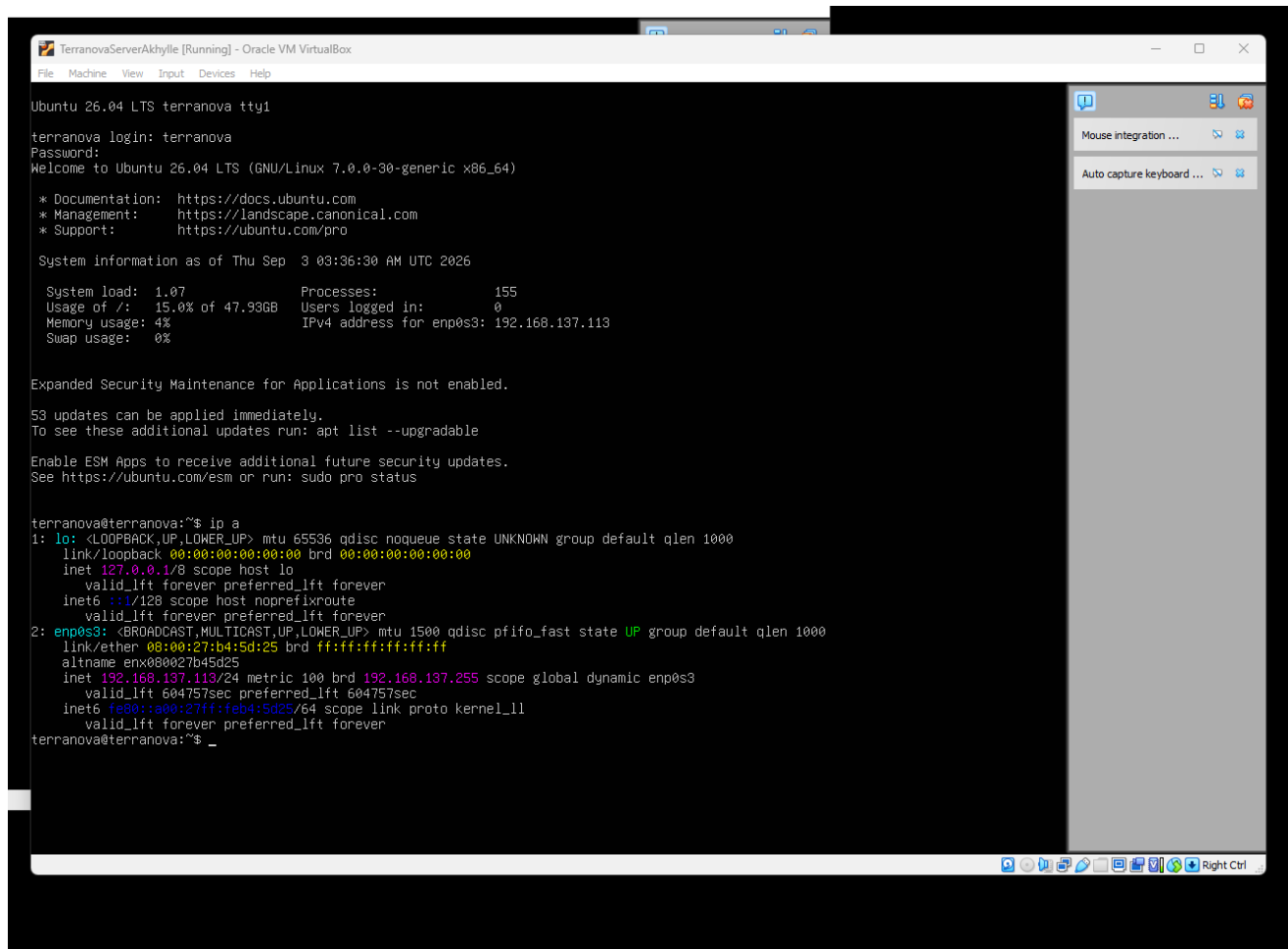
Error: Unable to locate package openssh-server-y
terranova@terranova:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 58
terranova@terranova:~$
```

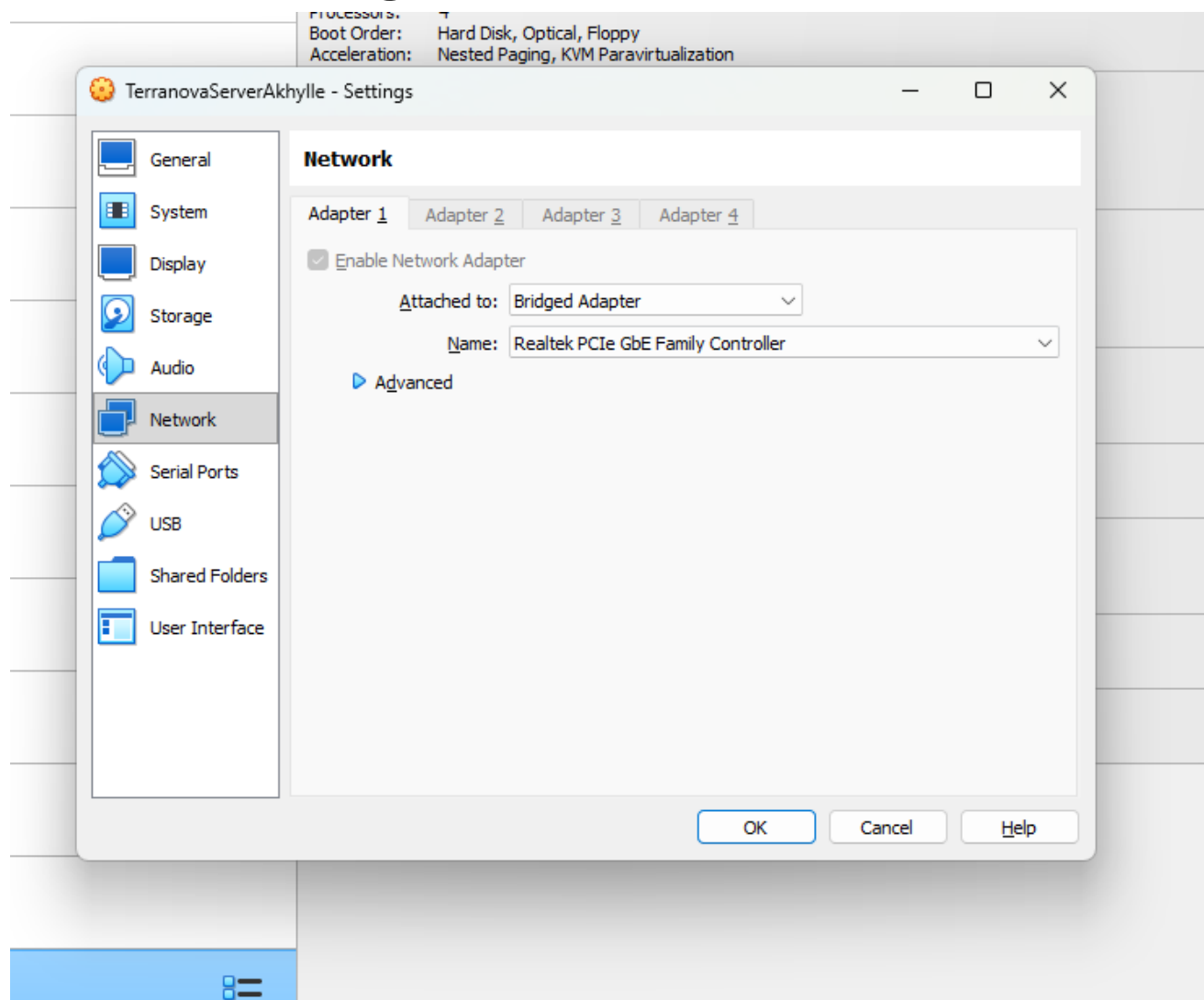
```

TerranovaServerAkhyll [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Err:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
  Temporary failure resolving 'ph.archive.ubuntu.com'
58 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Temporary failure resolving 'security.ubuntu.com'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
terranova@terranova:~$ sudo apt install openssh-server-y
Error: Unable to locate package openssh-server-y
terranova@terranova:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 58
terranova@terranova:~$
a      install      openssh-server  ssh      sudo      update
apt     ip      openssh-server-y  status   systemctl  -y
terranova@terranova:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' + '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' + '/usr/lib/systemd/system/ssh.service'.
terranova@terranova:~$ sudo systemctl start ssh
terranova@terranova:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b4:5d:25 brd ff:ff:ff:ff:ff:ff
    altname enx000027b45d25
    inet6 fe80::a00:27ff:feb4:5d25/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
terranova@terranova:~$
```



Network Configuration



Desktop Connection to the Server

```
terrano@192.168.137.113: Permission denied (publickey,password).
terrano@terrano-VirtualBox:~$ ssh terrano@192.168.137.113
terrano@192.168.137.113's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 03:39:41 AM UTC 2026

System load:  0.06               Processes:            159
Usage of /:   15.0% of 47.93GB   Users logged in:     0
Memory usage: 5%                IPv4 address for enp0s3: 192.168.137.113
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

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